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Frame Sequencing the Climate Change Online: An Analysis of Evolving Narratives on YouTube Video Data

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Abstract

Over the past decades, YouTube has become a central arena for public narratives where global risks, such as climate change, are increasingly framed for online audiences. We present empirical research conducted through a novel Framing Analysis approach, which we define "Frame Sequencing", aimed at analysing large volumes of unstructured data, particularly valuable for qualitative and quantitative inquiries on audiovisual content online. Methodologically, we offer a replicable workflow aimed at a robust Framing Analysis on user-generated video data, showing how context evolution shapes the prominence and articulation of online communication on the platform.

Finally, the proposed framework has been proven on YouTube platform following the discussed issue of climate change, where it insightfully highlighted the frame sequences on which a qualitative deepening has been carried out, leading to the acknowledgment of three evolving narratives portrayed by influential platform actors: Data room narratives; Science and call to action; and climate change political debate.

Keywords: Framing Analysis (1); Methodology (2); Platform Narratives (3); Video data (4); Climate Change (5)

1 Introduction

YouTube has become a central arena for public narratives about climate change, where content creation can be used as a form of audiovisual Framing to shape how global risks and debates are explained and thematised to the general audience online (Gamson and Modigliani 1989; Carvalho 2010;

Schäfer and Schlichting 2014; Dudo and Atkinson 2017). On YouTube, issues such as climate change are made visible, interpreted and negotiated through audiovisual narratives that combine scientific information, aesthetic choices and engagement-oriented formats (Dudo and Atkinson 2017; León et al. 2018). Nevertheless, most studies of climate change communication on social media still privilege posts and comments for analysis or focus on narrower contexts (Schäfer and Schlichting 2014; Pearce et al. 2019), leaving unexplored how Framing practices evolve over time, especially on platforms that require their content creators to develop specific strategies for content production (León et al. 2018; Reynolds and Hallinan 2024; Scharlach et al. 2024). On YouTube, climate change-related content is not simply presented as static information: instead, it emerges from ongoing negotiations regarding what constitutes relevant evidence, credible expertise, and legitimate definitions of the problem. Literature recognises this process as Framing (Entman 1993; Nisbet 2009; Carvalho 2010). Since video data has proven a valuable resource in qualitative research (Wang and Lien 2013), we conducted a longitudinal Framing Analysis of the platform, as it effectively captures a diverse range of User Generated Content (UGC) created not only by end-users but also by media organisations, independent experts, and media activists.

Previous research on Framing Analysis has shown how public issues are organised through configurations that select certain aspects of reality, connect them in causal links chains and make particular solutions appear more or less appropriate based on context (Goffman 1974; Entman 2003; Pan and Kosicki 1993; Scheufele 1999), then used to influence other journalists and even governments into specific agendas (Riker 1986).

In the field of climate change communication, these configurations influence how risks are perceived, how responsibility is attributed and how mitigation or adaptation pathways are imagined (Carvalho 2010; Schäfer and Schlichting 2014; Guenther et al. 2023). On YouTube, such frames are articulated over time through verbal argumentation as well as via visual repertoires, editing choices, tone of voice and interaction with audiences in comment sections, making the platform a key site for observing how climate change is articulately constructed as a public problem in the digital mediascape (Burgess and Green 2018; León et al. 2018).

Considering that "Framing provides the context that activates prior knowledge" (Cappella and Jamieson 1997) and newsmakers organise frames into the so-called "trains of thoughts", where context emerges over time (Price et al. 1997), we structured a longitudinal Framing Analysis to achieve

the most nuanced representation of frames over time while providing a solid base for further interpretations.

This temporal frame also presents substantive contextual relevance because it coincides with a period marked by COVID-19 and accelerated shifts in public discourse, during which climate change was publicly discussed and meaningfully re-articulated. During this phase, climate change communication circulated within a broader agenda shaped by overlapping crises, constituting a salient issue environment (Klenert et al. 2020; Zang et al. 2021). Additionally, the U.S. presidential election in 2017 and 2024 formed part of the contemporaneous context within which climate debates circulated in international media. Rather than estimating event-level effects, we interpret observed longitudinal shifts in YouTube climate narratives against this documented socio-political backdrop (De Pryck and Gemenne 2017; Sun and Ye 2025).

Therefore, we selected English-language videos from October 2020 to October 2025 as the temporal range for corpus construction. A five-year interval is widely recognised as a suitable timeframe for longitudinal analyses in media and communication research, as it allows the identification of both structural patterns and short-term discursive fluctuations, while remaining sufficiently contained to ensure analytical coherence (Saldaña 2003; Lüders et al. 2010; Rogers 2013).

Using video captions as a textual proxy for audiovisual content (Li et al. 2025), this research combines distributional semantics (Lenci and Sahlgren 2023) and hierarchical clustering with in-depth qualitative video analysis on an empirically informed subsample. Positioned within longitudinal scholarship on frame dynamics and temporal variation, our approach complements established efforts to track how frames change over time by moving from predefined frame categories to inductively detected trajectories grounded in platform data (de Vreese 2005; Matthes 2009). This novel approach, which we refer to as "Frame Sequencing" (defined in section §3.1), integrates distributional semantics with close qualitative examination, enabling the identification of temporal regularities that neither a purely computational analysis nor a purely qualitative reading would reveal on its own. In doing so, it also speaks to research on issue attention and public problem competition by treating shifts in narrative organisation as occurring within an evolving issue environment rather than as isolated textual fluctuations (Hilgartner & Bosk 1988). The sequential patterns detected through this combined procedure emerge only when lexical dynamics are examined across time and then interpreted through detailed inspection of the

underlying audiovisual content. The results show how the interplay between socio-political dynamics and platform logics contributes to reshaping the boundaries between descriptive, mobilising and agonistic forms of climate change communication (Gillespie 2010; Scharlach et al. 2024). These dynamics are thus analysed as platform-shaped processes, as YouTube's infrastructural conditions influence what becomes legible, comparable, and visible over time (Helmond 2015; Plantin et al. 2018).

The article is structured as follows. The first section presents Framing Analysis and situates YouTube climate change narratives within the broader socio-political and media context of the 2020-2025 period; the second one presents the methodological design; the third section reports the main findings, introducing the concept of "Frame Sequencing" and reconstructing the three that were found and their evolution over time. The discussion examines the results, highlighting the interrelationship between climate change communication and contemporary socio-political debates, while the conclusion outlines the main implications for future studies on climate change communication on YouTube.

2 Framing Analysis on YouTube, an ever-changing context

In the now classic text, "Frame Analysis: An Essay on the Organization of Experience" (1974), Erving Goffman reflects on the question: «under what circumstances do we believe things to be real?», borrowing the term "frame" from Bateson (1972), who uses it to describe the "serious" situations that arise during fantasy games, and which the author defines as "parentheses" within which something is real for those who participate. Aiming at a more operative definition, Entman (1993), following Fiske and Taylor (1991), clarifies that: «frames highlight certain bits of information about a communicated object, increasing its salience». The word "salience" means making a piece of information more recognisable, significant, and memorable to the audience. An increase in salience improves the likelihood that receivers will perceive the information, understand and process its meaning and store it in memory (Entman 1991). The temporal dimension is also present in the work of Reese *et al*. (2001), who look at frames as organising principles that are socially shared and enduring over time, able to work symbolically and significantly structure the social world.

On the other hand, YouTube has become one of the most vibrant environments for bringing social and ethical issues to global attention, not only as a participatory space but as a socio-technical system where ranking metrics, content moderation, and creator-economy incentives condition what circulates and how it is formatted (Gillespie 2010; van Dijck et al. 2018; Bucher 2018; Cunningham and Craig, 2019). Issues related to healthcare (Johnson *et al*. 2019), fake news and hate speech (Lee et al. 2021; Sauntson 2025) geopolitical tensions and propaganda (Dekker and Nguyen 2020) among others, have been discussed and negotiated on the platform over the years (Lewis and Christin 2022).

There is also a vast amount of research techniques for carrying out a Framing Analysis on legacy media, many of which are summarised in a literature review by Ali and Hassan (2022), and others of more recent conception (Hamborg 2023), but carrying out an analysis on YouTube means both considering its main characteristics and adapting the analysis techniques to an ever-changing environment.

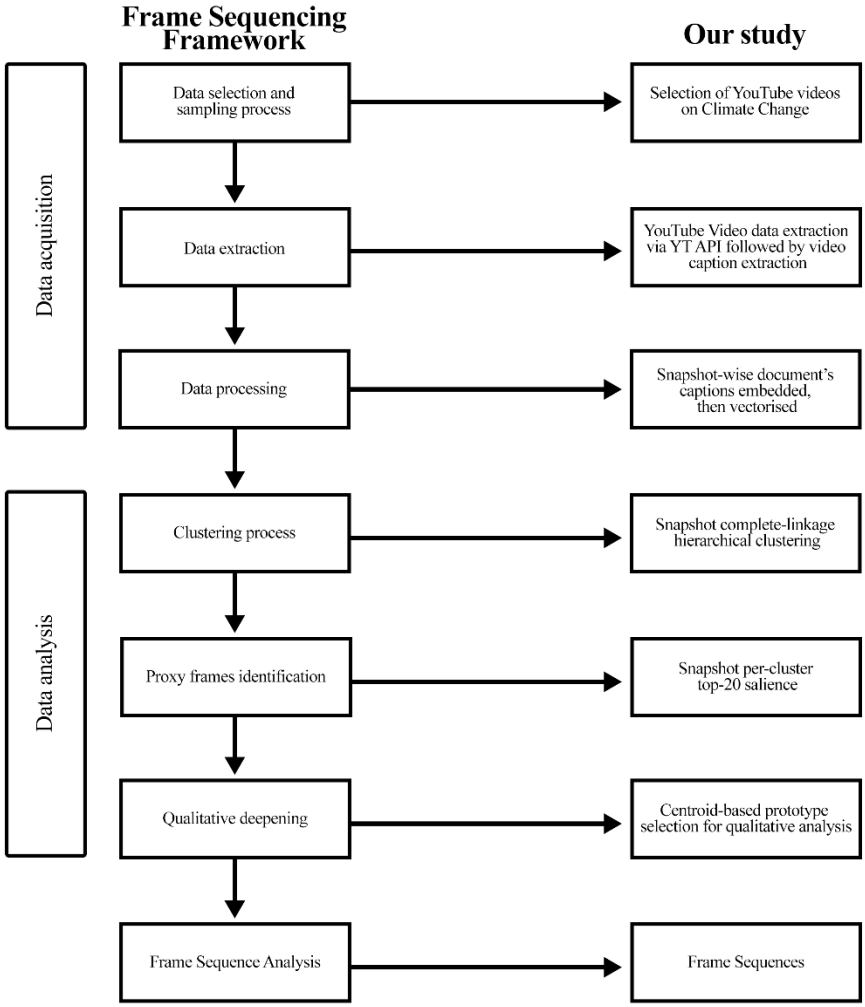
Among the main differences, of course, is the audiovisual dimension, with meanings articulated by images and sounds that would be impossible to present in legacy media (at least in the same quantity), a phenomenon reported as multimodality which «involves complex uses of spoken language, written language, visualisations in diagrammatic and pictorial form (increasingly often dynamic), as well as forms of metacommunication (spoken pointers, manual gestures, dynamic presentation resources such as zooms, slide transitions, gradual text reveals) that explicitly synchronises the diverse combinations employed» (Bateman et al. 2021). Although analysing captions means working on texts, these derive from transcriptions of dialogues that transform originally multimodal communicative practices into written artefacts, which differ from traditional textual production (Ochs 1979; Bucholtz 2000; Flewitt et al. 2009; Mondada 2018; Dancygier 2023) and require dedicated analysis. Among the similarities, however, is one of

the most important factors we mentioned earlier: the temporal axis. On YouTube, temporality is also infrastructural, because visibility is continuously recalibrated through ranking, recency signals, and recommendation, which can amplify or dampen narrative formats over time (Bucher 2018; Covington et al. 2016). According to the literature reported so far, a framing attempt must be continuous over time, making the user fill the empty "slots" with information on a given subject, thus updating their knowledge (van Dijk, Kintsch 1983). As in Scheufele (2004) «For example, someone who reads an article about the situation in the Middle East has already formed an idea of the situation. He will change it, however, refining and redrawing it on the basis of the information contained in the article». When addressing phenomena such as climate change, this calls for a longitudinal framing analysis that tracks how narratives evolve, especially during significant transformations. This approach should also produce results that can be examined through in-depth qualitative methods. The methodological choices that support this research design are outlined in the following section.

3 Research design

The main research questions that guided this study are: how global climate change communication on YouTube has evolved between 2020 and 2025, and whether, over this time span, some framing strategies have been continuously reproduced or reformulated. To address these research questions, we conducted an empirical study using a novel approach to analyse large volumes of unstructured YouTube data on climate change communication, applying it to a five-year corpus of sufficiently long videos to advance framing analysis techniques for audiovisual content online. For each video, we extracted the captions and analysed them using a computational method, followed by a subsequent qualitative deepening.
Figure 1 summarises the framework workflow presenting the proposed approach on the left and our study on the right.

Figure 1 - Framing sequencing research framework workflow



The corpus was constructed through systematic daily queries between October 1st, 2020, and October 1st, 2025, to the YouTube API with the keyword "climate change". This strategy approximates a search-surfaced issue space rather than an exhaustive archive of all climate-related uploads, as results may be shaped by query design, indexing practices, and undocumented changes in search ranking and relevance criteria over time. The initial extraction yielded more than 50,000 videos. From this substantial dataset, we retained only videos that met three selection criteria: English-

language captions[1], at least one user comment, and an explicit indication of English as the only admitted language. More than a mere "subset", the corpus retrieved was intended as a platform-conditioned population produced by affordances and unequal conditions of legibility and participation: caption availability and "commentability" are unevenly distributed across channels, therefore shaping which climate narratives become computationally processable and socially responsive within the platform's public space (Helmond 2015; van Dijck et al. 2018; Gillespie 2018). Applying these filters resulted in a dataset of 1,729 videos. To reflect platform logics, we distinguished short videos (≤180s) from long videos (>180s) and focused the analysis on the long-form videos (N=1,171).

Short videos were treated as a separate category and excluded from the analysis. This decision reflects a precise methodological consideration: because framing analysis requires sufficient discursive material to identify a coherent configuration of framing functions, highly compressed content often under-specify these elements and can reduce the validity and stability of frame inferences (Entman 1993; Matthes and Kohring 2008; Krippendorff 2018). In the YouTube context, short-form videos follow platform-specific logics, rapid consumption, compressed argumentation, and distinct aesthetics that differ substantially from long-form videos (Chen et al. 2024). Accordingly, we treat the exclusion of short-form content not only as a scope delimitation but as an analytic distinction between format regimes. Long-form videos enable extended evidentiary narration and cumulative framing work (De Lara et al. 2017), whereas short-form videos are typically optimized for high-velocity attention and different conventions of visibility, circulation, and engagement (Chen et al. 2024; Zulli and Zulli 2020). Integrating both formats would therefore conflate distinct platform grammars rather than increase corpus coverage under comparable analytical conditions.

No further exclusions were applied beyond the criteria described above, and the final dataset therefore reflects the sample of climate-change-related videos meeting these platform-derived conditions. Captions were gathered through a third-party provider that utilises the YouTube API to extract them

[1] This criterion, which seems to limit the selection, does not significantly reduce the sample size because YouTube automatically generates captions for videos uploaded to the platform. However, our research identified instances in which captions were generated in unrecognised characters or content creators voluntarily removed them from their videos. The 24 videos that fell into these categories were excluded from the analysis and are not part of the selected sample.

in bulk. After this process, we obtained a corpus subdivided into annual timeframes, organised into five "snapshots" (Rossetti and Cazabet 2018).

The analysis was threefold. First, we pre-processed the text and obtained vector representations of the video captions using a pre-trained FastText model (Grave et al. 2018). Captions were represented through mean pooling of word vectors to obtain a document-level embedding for each video. We then computed pairwise cosine distances on these embeddings and used the resulting distance matrix as input for a hierarchical agglomerative clustering with complete linkage (Murtagh and Contreras 2012). We selected this method because cosine-based complete linkage operates directly on the distance matrix and does not rely on additional metric assumptions. This property makes it suitable for high-dimensional representations such as FastText embeddings (Cozzolino 2022; Sinoara et al. 2019; Ostendorff et al. 2022). Compared to single-linkage, complete linkage avoids chaining effects generated by local similarities and tends to produce more internally coherent clusters (Murtagh and Contreras 2012). While this step provides a reproducible map of distributional similarity in transcribed speech, we do not assume that semantic proximity is isomorphic to frame similarity; rather, clustering is treated as an exploratory approximation of latent similarity in lexicalised narrative registers (Curran 2004), whose interpretation as framing is established through subsequent theory-driven qualitative validation (Entman 1993; Matthes and Kohring 2008). To determine the dendrogram cut for each yearly snapshot, we applied a fixed cut at 80% of the maximum dendrogram height, so that the number of clusters k is an outcome of this threshold rather than a parameter tuned ad hoc, yielding:

$$k_{2020-2021} = 3,\ k_{2021-2022} = 5,\ k_{2022-2023} = 3,$$
$$k_{2023-2024} = 3, \text{ and } k_{2024-2025} = 4$$ [2]

We did not use internal separation indices as decisive criteria for cluster "quality", because porous semantic boundaries and overlap are theoretically plausible properties of public discourse and often reduce geometric separability in embedding spaces (Rosch 1999; Sharma et al. 2023); instead, clusters are used here as a reproducible partitioning device to support the

[2] We operationalised time through five annual snapshots defined as consecutive October–October windows (1 Oct 2020–1 Oct 2021 till 1 Oct 2024–1 Oct 2025), and we report cluster counts accordingly.

identification of candidate lexical regularities and the subsequent qualitative reconstruction of framing sequences (Macia 2015).

Second, we examined the clusters for each year and identified the 20 most frequent words within these clusters, treating this frequency list as a descriptive heuristic to support visualisation and preliminary orientation rather than as a direct proxy for frames (Figure 2, see Findings), and using these terms to initially track diachronic lexical salience.

Third, we identified prototype videos by computing cosine distance from each cluster centroid and selecting the three closest videos per cluster, yielding a total of 54 prototype videos for subsequent qualitative deepening through iterative viewing, note-taking, and the compilation of analytic memos (Kalogeratos and Likas 2011). Selecting a small set of centroid-nearest cases per cluster was intended to balance coverage of cluster variation with the practical constraints of in-depth audiovisual interpretation (Tracy 2010). The interpretive process followed a systematic, iterative approach in which we compared our observations across videos and clusters. Although the study did not employ a formal multi-coder protocol, the analytical categories were refined through internal cross-checking and discussion among the authors to enhance interpretive consistency, and we documented this process through a shared analytical template and an audit trail to improve transparency and traceability of interpretive decisions. The combination of descriptive salience profiles and close examination of the prototype videos supported a more robust interpretation of the clusters and the detection of recurrent Framing Sequences. The sole computational analysis would have yielded clusters with limited capacity to account for the underlying phenomena. At the same time, the sole qualitative approach could not capture the quantity of videos we selected. The triangulated procedure offered analytical depth that neither approach could achieve on its own, enabling a deeper detection of recurring framing sequences in climate change narratives on YouTube, which is the focus of the subsequent sections of this article.

4 Findings

4.1 Introducing the frame sequencing

Here we introduce the concept of “Frame sequencing”, a novel Framing Analysis approach which involves identifying coherent narrative segments and their patterned ordering as they emerge and evolve during the selected

timespan. Therefore, in this study we use the term Framing Sequences to denote dynamic configurations that trace the temporal unfolding of framing practices across the five-year corpus. Specifically, while a "frame" refers to a relatively stable pattern to suggest a problem definition, causal interpretation, and normative orientation (Entman 1993; Scheufele 1999), a Framing Sequence captures how such patterns recur, mutate, or realign over time within the platform's evolving discursive environment. The analytical need for a "sequence" construct arises when longitudinal change is not adequately described as variation in frame prevalence alone, but as the re-composition of framing contents into recurrent temporal arrangements, such as stable problem presentations, recurring evidentiary segments, shifting attributions of responsibility, and the periodic reappearance of action-oriented closures. In this sense, "sequence" is a higher-order construct with three minimal properties: *recurrence*, the same functional arrangement reappears across snapshots; *mutability*, the arrangement persists while internal elements shift in emphasis or lexicon; and *realignment*, the arrangement changes in response to contextual and platform conditions, producing detectable re-orderings of framing functions.

The focus is not only on which frames are present, but on how framing is assembled and re-assembled over time through relatively recognisable narrative configurations. It is also complementary to research on issue-attention cycles (Downs 2016), because it treats attention shifts as conditions under which particular framing arrangements become more likely to stabilise, fragment, or polarise, rather than as explanations of content change per se (Hilgartner and Bosk 1988; Matthes,2009).

Operationally, this approach differs from existing longitudinal framing studies in news media (de Vreese 2005; Matthes 2009; Cobo et al. 2025), which typically track predefined frame categories over time. Here, sequences emerge inductively from a combination of distributional patterns and qualitative interpretation, allowing the analysis to capture platform-specific dynamics, such as creator practices, audiovisual conventions and algorithmically shaped visibility, that play a constitutive role in how climate narratives evolve on YouTube. To qualify a pattern as a Framing Sequence in this study, we required converging evidence across three layers: repeated cluster-level lexical signatures across multiple snapshots (e.g., recurring salient term profiles rather than one-off topical spikes); consistent multimodal-narrative organisation in the prototype videos (e.g., comparable ordering of framing functions and recurrent presentational conventions); and an interpretable mechanism linking the observed persistence or shift to

platform and socio-political conditions (e.g., changes in creator strategies, genre conventions, or visibility dynamics).
Our Frame Sequencing proposes a novel way to theorise longitudinal framing on platformised media as the evolution of recurrent narrative arrangements are shaped by both issue environments and platform logics, rather than as a mere rise-and-fall of issues.

4.2 Results

We enriched the computational analysis of the most frequent words in each cluster with a qualitative interpretation based on an in-depth inspection of the 54 videos that emerged. We tracked the longitudinal development of the terms associated with each cluster, emphasising those that frequently co-occurred over time or shared conceptual linkages, which contributed to defining each Framing Sequence. We treated terms from both higher-order and lower-order categories, as well as lexical variations and changes in discursive roles, as parts of the same underlying narrative sequence.

For visualisation purposes, we limited the graphical representation to the twenty most frequent terms per cluster, out of the over a hundred proposed. The result presented in Figure 2 is based on this analytical process, using the twenty most frequent terms as guidelines. The colour distinctions that differentiate the framing sequences were conceptually defined only after the in-depth inspection of the selected videos.

Based on the aforementioned research approach, the analysis of the five-year snapshots reveals three recurring Framing sequences that shape how climate change is narrated and addressed on the platform. These sequences are: (a) “Data Room Narratives”, (b) “Science and Call to Action: Civic Mobilisation”, and (c) “Climate Change Debate: Crisis, Adaptation, and Answers”. These are not static categories, but dynamic narrative structures: their prominence, internal structure, and degree of overlapping terms change over time. This evolution reflects broader shifts in public attention, political context, and platform dynamics. Each sequence arises from specific lexical trends identified through clustering and from the narrative, argumentative, and performative characteristics observed in the qualitative subsample.

Taken together, these three framing sequences provide a coherent analytical map for understanding how climate change is discursively organised on YouTube over time for the Anglophone public. The framing sequences allowed the qualitative reconstruction of the dominant narrative trajectories and the subtle shifts in emphasis, alignment, and conflict that

characterise the platform's evolving discourse. This framework serves as the basis for the detailed examination presented in the following sections, where each framing sequence is analysed in depth.

Figure 2 - Framing sequences over time

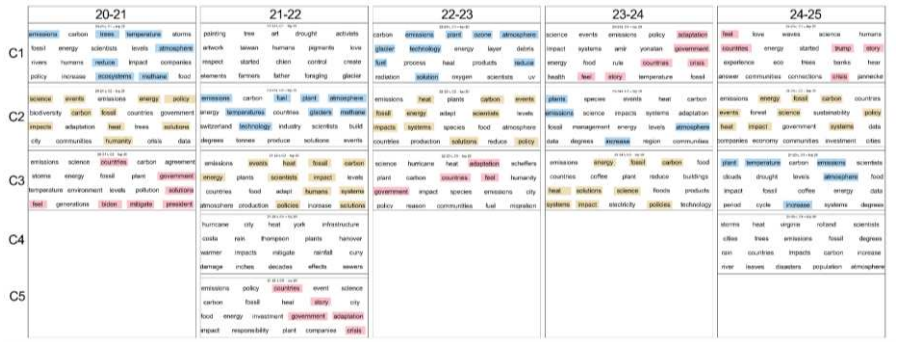


Figure 2 presents the diachronic evolution of the three framing sequences identified in the analysis (by colour) using the twenty most frequent terms for each cluster, distributed across the five snapshots from 2020 to 2025. The figure is organised into cluster rows and snapshot columns, enabling us to observe how the key lexical elements present in the clusters constituted the three framing sequences that change, strengthen, or weaken over time. The figure also highlights the appearance of certain "bridge" words across different snapshots, indicating moments of convergence or shifts in discourse.

The first framing sequence, coloured blue, which we have labelled "Data Room Narratives", highlights a lexicon that remains anchored to technical-scientific semantic fields with keywords such as "emissions", "carbon", "atmosphere", "temperature", "ozone", "glaciers", "plant", "fuel", and "methane". This sequence maintains coherence and continuity throughout the entire timespan, indicating the persistence of a narrative centred on data, evidence, and biochemical processes, with few entries featuring strong or politically connotative terms.

The second framing sequence, in yellow, labelled "Science and Call to Action: Civic Mobilisation", instead displays a semantics oriented toward the relationship between science and public action, with terms such as "solutions", "impacts", "systems", "heat", "policy", "adaptation", "science", "management", and "sustainability". In this sequence, the development reveals a progressive stabilisation of a discursive repertoire linking solution

engineering, risk management, and governance elements. This framing sequence highlights how, from 2022 onward, terms referring to collective action, management, policies, and systems became more frequent and systematic, giving a greater contribution to the structuring of the framing narrative.

The third framing sequence, coloured pink, which we have named "Climate Change Debate: Crisis, Adaptation, and Answers", traces a more irregular and politically charged trajectory. Terms such as "countries", "government", "president", "crisis", "adaptation", "migration", "feel", "story", "Biden", "Trump", and "humanity" emerge. In this sequence, the figure shows how the climate debate, increasingly polarised, incorporates political and institutional elements, subjective and emotional narratives, and more controversial issues such as migration, responsibility, and political leadership. The uneven pattern of terms reflects the dialectic nature of this framing sequence, which intensifies during periods of crisis, extreme events, or discursive conflict.

Overall, our figure, while representing an overview of the process, makes visible the intersection between lexicon, temporality, and discursive configurations. The result is a diachronic framework that allows us to read the transformation in the ways YouTube frames climate change through three distinct but intertwined narratives.

4.3 Data Room Narratives

The first sequence is characterised by a data-driven communication style that focuses on providing information mostly drawn from climate change reports, in a documentary style. Videos in this category function as "data rooms"[3]: they reconstruct climate change through the visual grammar of expert briefings, combining scientific reports, geophysical modelling, satellite imagery and 3D graphic representations of the Earth system. This register foregrounds not only the authority of data but also its performative staging, in which the video makers adopt visual conventions reminiscent of documentary science communication, repeatedly anchoring the narrative in technical evidence. As a result, climate change is framed primarily as a

[3] "Data room" journalism refers to the use of large data sets in the news industry, a field also known as data journalism.

quantifiable, measurable and model-based phenomenon, whose risks can be understood by mastering the underlying data.

Within this frame, content creators tend to position themselves as intermediaries of scientific expertise, employing a deductive architecture that moves from the planetary scale to case-specific implications. The narrative rhythm is linear and didactic, emphasising clarity, step-by-step explanation and an impersonal tone. Climate change is thus framed less as a contested political issue and more as a technical problem to be correctly represented.

4.4 Science and Call to Action: Civic Mobilisation

The second framing sequence is centred on mobilisation. In these videos, scientific knowledge is reinterpreted through a pragmatic, future-oriented lens, with the purpose not simply to describe the climate crisis but to translate understanding into action. The framing logic is moving from evidence to implication and from knowledge to commitment. This communication style became evident in the 2021-2022 phase, when the pandemic context amplified the connection between scientific communication and civic mobilisation, and much of the climate change conversation circulated through online conferences, remote seminars and digital public events. During that period, the online environment, urging individuals to take concrete measures to protect their health, also framed how the climate change narrative was communicated, spurring mobilisation, partly due to signs of environmental improvement stemming from near-zero use of fossil fuels. This call-to-action framing, although it has seen some shifts in the subsequent years, has remained stable in its communicative style.

In this register, content creators integrated expert input with explicit invitations to act, ranging from “how-to” guides on sustainable practices to broader appeals for collective responsibility. The narrative architecture tends to follow a problem–solution pattern, in which a scientific exposition is immediately followed by a set of actionable pathways, often illustrated through community initiatives, policy proposals, or technological innovations. Audiovisually, these videos assume a more relational tone by directly addressing, with emotionally coded pacing that establishes a sense of shared purpose. The videos may also incorporate fragments of online events (such as conferences, panel discussions, and webinars) that reflect a moment when the boundaries between scientific dissemination, advocacy, and civic education were unusually permeable.

In the last snapshots, this framing sequence becomes more overtly political: initially, the videos tend to emphasise individual responsibility and possible behavioural adjustments, but later ones begin to situate climate action within debates about justice, governance, and inequality. Therefore, the framing evolves from the appeals to reduce emissions or adopt sustainable habits to the reflections on institutional accountability and the uneven distribution of climate risks. This shift mirrors a broader transformation in public discourse. As the socio-political implications of climate change become more salient and the pandemic's communicative environment recedes, mobilisation is no longer framed merely as a personal choice but as a collective task requiring coordinated social and political intervention. In this sense, the framing sequence captures the transition from an early phase in which civic appeals emerge organically within digital scientific spaces to a later phase where calls to action articulate a more explicit theory of conflict and institutional change.

4.5 Climate Change Debate: Crisis, Adaptation, and Answers

The third framing sequence is defined by explicit instances of confrontation and by overtly argumentative, polemical, or politically charged content. These videos do not merely present information but actively enter the arena of climate change debate, addressing and replying to denialist claims spreading, and tackling narratives that minimise the severity of the crisis. Rather than focusing on technical explanation or mobilisation, they articulate climate change as a discursive battleground structured by conflict, disagreement and competition for epistemic authority.

This register becomes increasingly polarised as the snapshots move to the later years, reflecting the progressive politicisation of climate change discourse, particularly in the United States, from the period marked by the 2022 midterm elections to the 2024 presidential campaign. The videos' communication style is structured after talk shows, online political commentary, or podcast-style debates, with a multi-voice format featuring hosts, commentators, and recurring ideological antagonists. In this setting, climate change is addressed less as a scientific object and more as a proxy for broader political identities, cultural alignments and partisan conflicts. As these formats invade the YouTube climate space, the resulting discourse becomes more agonistic, with content creators positioning themselves as

participants in a contested public sphere rather than neutral disseminators of information.

Rhetorically, videos in this sequence rely on a dialectical approach, in which claims are introduced, challenged, reframed and contrasted. Content creators often alternate between confutation and demonstration, combining selective data with narrative tactics designed to expose inconsistencies, reveal logical fallacies or highlight the political motivations underlying sceptical positions. This style oscillates between patient explanation and openly confrontational commentary, depending on the content creator's persona and intended audience. Crucially, denialist narratives are not simply dismissed but used instead as discursive catalysts through which creators reaffirm scientific legitimacy or articulate concerns about institutional responsibility, political obstruction, or the consequences of policy inaction.

The expansion of this framing sequence over the snapshots is not merely a by-product of platform dynamics but reflects a broader historical and political shift. In this framing sequence, the discursive emphasis shifts from the risks documented by scientific evidence to those generated by political inaction, partisan obstruction, and strategic misinformation. Likewise, here the question of responsibility is reframed, as it no longer concerns only individual behaviours or national emissions trajectories, but also the actors who shape public understanding through conflictual narratives. The race to lead the climate change debate becomes a competition of imaginaries about the future, facing each other through narratives of catastrophe, denialism, resilience, and techno-solutionism.

5 Discussion

The evolution of the three framing sequences across the snapshots of the five-year corpus reflects a complex interplay between content creators' communication style and crucial socio-political and historical developments. This temporal span appears to coincide with a period marked by overlapping crises that repeatedly reshaped public agendas.

In the first framing sequence, the use of a scientific register reflects a communication style that resonates with audiences familiarised with documentary-like formats. In this sense, the risk is framed primarily as a physical, measurable phenomenon. At the same time, responsibility is associated with institutional and macro-level actors, and mitigation is seen as a technical pathway aligned with scientific feasibility.

The second framing sequence foregrounds the socio-political environment, which grew unstable, leading to the urge for mobilisation-oriented narratives. Online conferences, remote activism, and the revamping of public debates created favourable conditions for content that translated scientific insights into behavioural cues, ethical appeals, and visions of collective responsibility. The framing of risk became more situated and socially grounded, connecting climatic impacts to inequalities, community vulnerability and intergenerational concerns. Mitigation, in turn, was reframed as a domain of civic engagement and moral agency rather than a purely technical trajectory.

The third framing sequence emphasises political contrasts that shifted the salience and emotional charge of climate change discourse, highlighting similarities between the analysed climate change communication and that observed in previous years (Marshall 2015). The 2024 U.S.A. presidential election added another layer of volatility in the observed debate, amplifying polarisation around environmental policy and reinforcing the visibility of antagonistic narratives. Responsibility shifted towards communicative behaviour and political obstruction, while mitigation became an arena of ideological dispute.

Taken together, the three framing sequences show the dynamic representation of climate change across YouTube, where different communication styles have been employed to address the convergence of scientific development, activism and geopolitical events.

This diachronic perspective underscores how climate change communication on YouTube continually adapts to the socio-political environment in which it is embedded, making the platform a key site for observing the reconfiguration of contemporary climate narratives.

6 Conclusion

The analysis results show how on YouTube thrives an evolving and heterogeneous discursive ecosystem in which climate change communication is articulated through shifting combinations of scientific explanation, civic mobilisation and public contestation.

The proposed methodological approach highlights the value of analysing climate narratives on YouTube through a sequential and temporally sensitive research design. The contribution of this approach is to reveal how frames evolve over time, making visible patterns of divergence and realignment that would remain hidden in cross-sectional analyses of climate communication.

By tracing the trajectory of these framing sequences across the snapshots, the study offers an empirically grounded account of how platform dynamics and socio-political events shape the evolution of climate-related discourses. The combination of hierarchical clustering on vectorised video captions and in-depth qualitative video analysis enables the reconstruction of narrative configurations that would be difficult to identify through a single analytic lens. Thus, given the volume of videos uploaded daily on these platforms, this approach shows a replicable framework for future studies seeking to examine the complexity of contemporary world communication within platformised media ecosystems (Mattes and Kohring 2008; Salganik 2019).

The multimodal nature of content on platform-based systems such as YouTube poses specific analytical challenges. Any attempt to study these videos must consider how spoken language, written captions, dynamic visualisations, gestures and meta-communicative cues work together (Flewitt et al. 2009; Bateman et al. 2021; Mondada 2018). This multimodality expands the expressive repertoire through which climate change can be framed, enabling narrative density and affective resonance that go beyond purely textual formats (Ochs 1979; Bucholtz 2000). At the same time, relying on captions for computational analysis introduces a structural reduction: captions transcribe spoken interactions but inevitably turn complex multimodal practices into linear textual artefacts (Ochs 1979; Bucholtz 2000; Flewitt et al. 2009; Mondada 2018). Recognising this tension is essential for interpreting our findings, as it highlights both the potential and the limits of studying climate change communication on a platform where meaning is produced through the interplay of multiple modalities and stakeholders.

For these reasons, our results show how climate change is framed within a specific sample of YouTube videos over time, under conditions of discursive, political and technological change. By tracing the emergence and transformation of the three framing sequences, the study sheds light on the processes through which climate change narratives are produced, circulated and disputed on this platform.

The three framing sequences identified in the study capture the main trajectories through which a diverse array of stakeholders renders the climate issue visible, intelligible and actionable for diverse audiences. Their evolution over time reflects the interaction between changes in the socio-political environment and the adaptive strategies of content producers. Our interpretation aligns with prior research on climate change framing in online video (León et al. 2018), which shows that framings are not randomly

distributed but patterned by institutional roles, professional routines, and broader political–economic logics that shape which narrative elements are foregrounded and which forms of mobilisation are encouraged. In this light, the presence of scientific–technical framing to mobilisation-oriented and polarised political framing does not merely reflect shifts in stylistic preferences: it mirrors a progressive reconfiguration of how different stakeholders seek to manage attention, authority and affect in a context of overlapping crises. Our three framing sequences can therefore be read as specific instantiations of dynamics already identified in the literature, where climate narratives move from informational and expert-led registers towards more affective, gain/loss-oriented and conflictual representations tightly coupled to institutional interests and partisan contestation (O'Neill and Nicholson-Cole 2009; Boykoff 2011; Nisbet 2009; Marshall 2015; León et al. 2018).

Future developments of this work could involve the short-video analysis, focusing on reels and shorts, a type of video format that is rapidly spreading across the principal digital platforms of the today mediascape, or considering video comments to study the interactive dynamics that arise between users and content creators, eventually employing fine-tuned computational approaches to study the visual aspects embedded in the images shown on screen.

Data availability

The data are available from the corresponding author on reasonable request.

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Declarations

Competing interests

The authors declare no competing interests.

Ethics declaration

Not applicable. The present study does not involve human participants, animals, or the use of sensitive personal data. No informed consent was required. The authors declare no competing interests and no specific funding sources related to this research.

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